Predicting Hotel Booking Cancellations with Machine Learning

A data-driven solution to help hotels identify high-risk reservations before arrival

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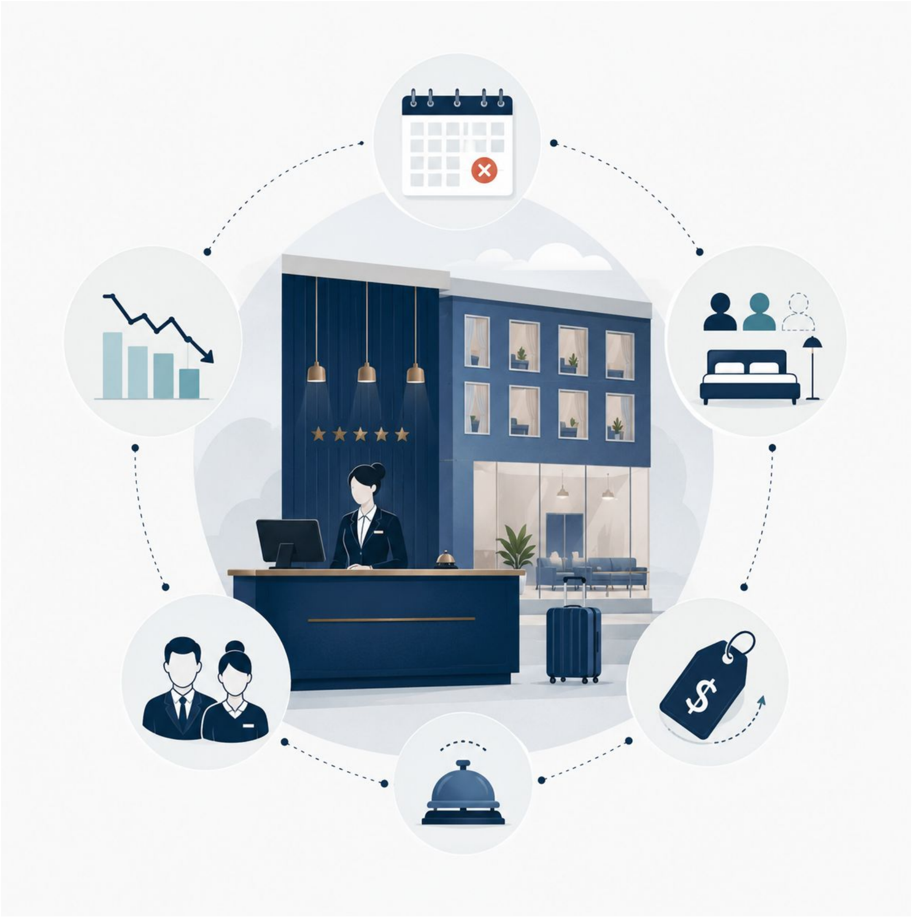
The Problem: Cancellations Create Uncertainty

Hotel cancellations affect more than one reservation.

They create uncertainty in:

- Revenue forecasting
- Occupancy planning
- Staffing decisions
- Pricing strategy
- Overbooking management

If hotels can identify high-risk bookings earlier, they can make better operational and commercial decisions.



From Reservation Data to Cancellation Prediction

Dataset:

Hotel Reservations Classification Dataset

36,275 hotel bookings

19 original variables

Target:
Booking_status

Not_Canceled: 67.2%

Canceled: 32.8%

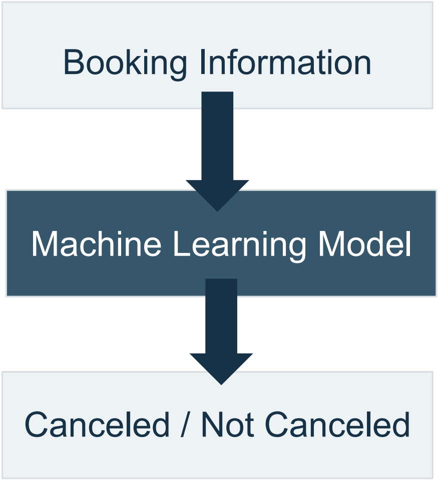
Moderately imbalanced

Prediction task:

Will this booking be cancelled?

Input
Booking details available before arrival

Output
Canceled / Not_Canceled



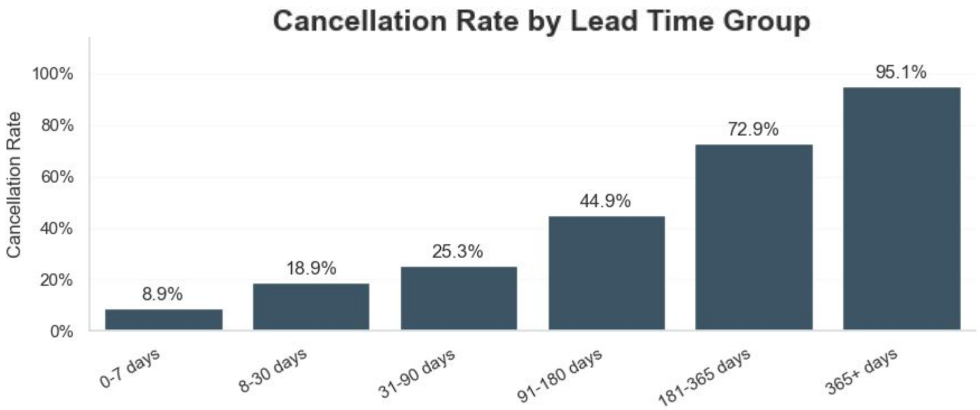
What Drives Booking Cancellations?

EDA revealed clear patterns across booking timing and guest commitment.

Lead Time

365+ days
95.1% cancellation rate

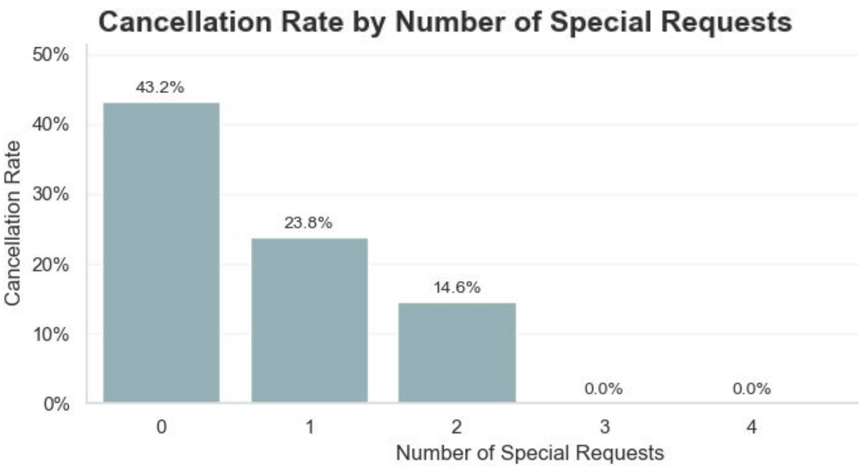
Bookings made further in advance are much more likely to cancel.



Special Requests

0 requests: 43.2%
2 requests: 14.6%

More special requests are linked to lower cancellation risk.



Cancellation risk is strongly linked to booking timing and guest commitment.

Building a Reliable Cancellation Model

All models were compared using the same preprocessing pipeline and test set.

1. Data Split

80% training / 20% testing

Stratified split to preserve cancellation balance

2. Preprocessing Pipeline

Numerical features → StandardScaler

Categorical features → OneHotEncoder

Booking_id removed

Fitting only with training data

3. Models Compared

- Logistic Regression
- Random Forest
- Gradient Boosting
- Decision Tree
- KNN Classifier

4. Evaluation & Optimization

Main metric: F1-score

Precision, recall and confusion matrix reviewed

GridSearchCV used for model tuning

Comparing Model Performance

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	0.9068	0.8882	0.8187	**0.8520**
Decision Tree Tuned	0.8841	0.8328	0.8086	0.8205
KNN Tuned	0.8816	0.8401	0.7888	0.8136
Decision Tree Baseline	0.8744	0.8109	0.8044	0.8076
KNN Baseline	0.8567	0.7996	0.7505	0.7743
Gradient Boosting	0.8584	0.8296	0.7148	0.7679
Logistic Regression	0.8143	0.7537	0.6437	0.6943

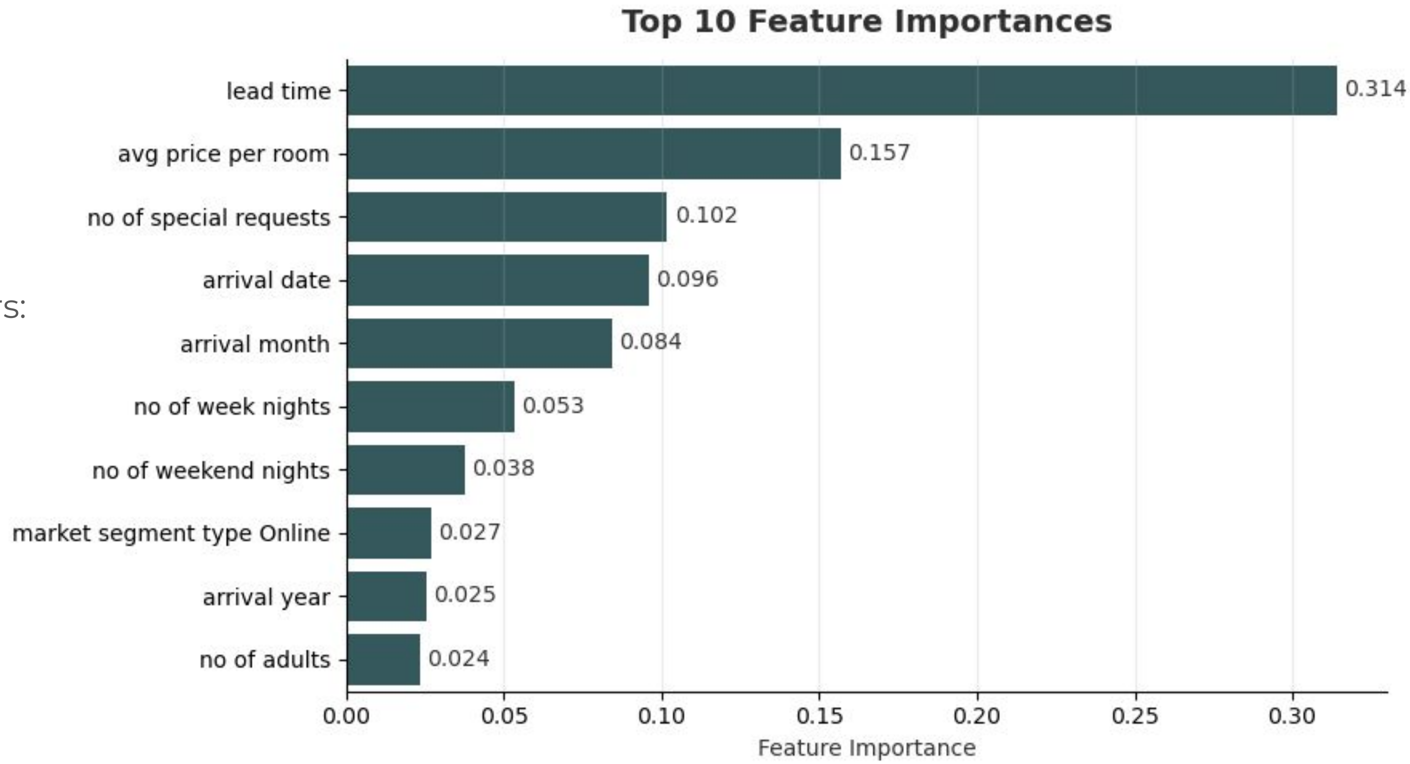
Random Forest delivered the strongest balance between detecting cancellations and avoiding false alerts.

Top Predictive Signals

Lead time dominates the model:
Bookings made further in advance carry much higher cancellation risk.

Guest commitment matters:
Special requests are an important signal of lower cancellation risk.

Price and timing add context:
Room price and arrival timing help refine the prediction.



The strongest model signals are consistent with the patterns found during EDA.

From Prediction to Hotel Decisions

Business Impact
How hotels can use StayGuard

- Flag high-risk reservations
- Improve occupancy planning
- Support overbooking decisions
- Prioritize guest communication
- Reduce revenue uncertainty

Limitations
What the model does not solve

- Predicts risk, not certainty
- No external data included
- Threshold depends on hotel strategy
- Results depend on dataset context

StayGuard gives hotel managers an early risk signal, not an automatic decision.

Final Takeaway

Cancellation risk is predictable when booking behavior is translated into data.

- Lead time: strongest cancellation signal
- Guest commitment: reduces cancellation risk
- Random Forest: delivered the best model performance

StayGuard helps hotels move from reacting to cancellations to anticipating them.

StayGuard

HOTEL DATA. SMARTER DECISIONS. SAFER REVENUE.

Thank You

Johannes Vidal

GitHub Repository:

github.com/JohannesVidal/stayguard-hotel-cancellation-prediction

Questions?